

KABILESH C

Software Engineer | Networking Protocols | Application Software
+91-9360267618 | kabileshchinnaswamy25@gmail.com | www.linkedin.com/in/kabilesh-c-1935a3262 | Coimbatore, India

PROFESSIONAL SUMMARY

Final-year ECE student with proven ability to develop software consistent with Design Thinking principles focusing on simplification, user privacy, and secure coding practices. Experienced in full SDLC, agile engineering, requirements analysis, design, coding, unit testing, and acceptance testing. Proficient in C, C++, and Python with hands-on expertise in networking protocols (TCP/IP, routing, switching), embedded Linux, OS fundamentals, and hardware/software interfaces. Strong collaborator with excellent communication skills and a self-motivated drive to achieve results.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer Intern | Emglitz Technologies

Jan – Feb 2025 | Coimbatore, India

- Developed software consistent with Design Thinking principles using secure coding practices on ESP8266/NodeMCU (C/C++); contributed to technical design of features and wrote clean, functional code with unit tests under supervision.
- Performed coding, debugging, testing, and troubleshooting throughout the development lifecycle; executed tests for scale, reliability, and velocity; monitored and triaged incident reports during platform bring-up.
- Collaborated with cross-functional teams across development and testing phases; estimated task completion times, communicated status and blockers, and identified paths to eliminate blockers and meet deadlines.
- Applied depth of knowledge in secure coding principles; reviewed code for compliance; shared knowledge from independent learning with the team to drive continuous improvement.

Software Testing Intern | Duratech Solutions

Jun – Jul 2024 | Coimbatore, India

- Participated in all SDLC phases requirements analysis, design, coding review, unit testing, acceptance testing, and release applying agile engineering practices; assisted in code reviews and addressed code review feedback.
- Executed API testing (k6) and load testing (Gatling) to validate performance, availability, reliability, scale, and velocity; analyzed defects, validated data mappings, and contributed to postmortem discussions.
- Established and sustained excellent relationships with extended team; collaborated effectively to share learnings, problem-solve, and deliver solutions that addressed customer needs.

PROJECTS

AI-IoT Smart Body Analyzer | Python, Flask, C++, Random Forest ML, ESP8266, I2C/SPI/GPIO, JavaScript

- Designed and built an AI-integrated IoT system on ESP8266 (NodeMCU) with I2C, SPI, GPIO interfaces; implemented low-level embedded software from technical design through completion demonstrating full ownership of a small-to-medium feature.
- Integrated Machine Learning (Random Forest) and built a Flask/JavaScript dashboard with RESTful APIs; applied understanding of how related features solve customer needs; scaled application to cloud infrastructure with performance optimization.

E-Commerce Platform – QuickBox | React.js, Spring Boot, MongoDB, RESTful APIs, Java

- Designed and developed a scalable full-stack application using object-oriented design; wrote clean functional code, unit tests, and conducted end-to-end debugging and troubleshooting from technical design through release.
- Built a React.js user interface demonstrating interest in UX/UI design; applied software development best practices for accessibility across devices, making technology accessible to people on a variety of platforms.

TECHNICAL SKILLS

Languages	C, C++, Python (strong), Java, JavaScript, Shell Scripting, SQL
Networking	TCP/IP Fundamentals, Routing & Switching Concepts, Network Debugging Tools & Techniques, Firewall, IPS/IDS, VPN, IPsec, TLS, Data Security, Threat Vectors
Embedded & OS	Embedded Linux, BSP, Boot-time Initialization, Memory Management, Interrupts, Concurrency, Device Drivers, Kernel/User Space, System Programming, UART, SPI, I2C, GPIO, ESP8266
Software Eng.	Design Thinking, Secure Coding, SDLC, Agile Engineering, OOP, DSA, System Design, Performance Optimization, Code Reviews, Debugging & Troubleshooting
Testing & QA	Unit Testing, Acceptance Testing, API Testing (k6), Load Testing (Gatling), Scale & Reliability Testing, CI, Defect Analysis, Postmortems, Incident Triage
Web & Tools	Spring Boot, Flask, React.js, RESTful APIs, MySQL, MongoDB, Linux, Git, VSCode, AI-Assisted Engineering, Open Source, Mobile App Development

EDUCATION

B.E – Electronics and Communication Engineering

2022 – 2026 (Expected)

Coimbatore Institute of Technology | CGPA: 7.8/10 | Coursework: Computer Networks (TCP/IP, Routing, Switching), OS, DSA, OOP, Embedded Systems, Microprocessors